INTRODUCTION

The 21-Day Programme is designed to help students improve their mathematical thinking, learn modern AI tools, work as a team, and develop practical problem-solving skills. During the first 10 days, we learned about mathematics, explored different AI tools, formed a team, selected a problem statement, and started developing a project using AI-assisted coding.

DAY-1 (MATHS AND ME)

On Day 1, we introduced ourselves and discussed our connection with mathematics. We shared our interests and understood how mathematics is useful in our education and everyday life.

Key Learning:
We understood that mathematics is not only a subject but also a tool for logical thinking and problem-solving.

—

DAY-2 (JUST A MINUTE)

On Day 2, we participated in mathematical activities that helped us use our mathematical knowledge quickly. We applied basic mathematical concepts and practiced thinking within a limited time.

Key Learning:
We improved our quick-thinking, calculation, and problem-solving abilities.

—

DAY-3 (NOTEBOOK LLM)

On Day 3, we learned how to use NotebookLM. We explored how AI can work with provided documents and information to help us understand, organize, and summarize content.

Key Learning:
We learned how AI can be used as a study and research assistant.

—

DAY-4 (NANO AI)

On Day 4, we learned about Nano AI/Nano Banana and explored how AI can be used for creative and visual tasks. We understood how prompts can be used to generate or modify visual content.

Key Learning:
We learned the importance of writing clear and effective prompts for AI tools.

DAY- 5 (CLAUDE AI ARTIFACTS)

On Day 5, we learned how to use Claude AI Artifacts. We explored how AI can help create useful interactive outputs and project-related content.

Key Learning:
We understood how AI can support the development of ideas, documents, and interactive project outputs.

DAY-6 (MATHEMATICS TEST)

On Day 6, we attended a test based on the mathematical concepts learned during the programme. The test helped us check our understanding and identify areas where we needed improvement.

Key Learning:
We evaluated our mathematical knowledge and improved our confidence in solving problems.

DAY-7 (SETUP)

On Day 7, we formed our project team and discussed different problems that could be solved using mathematics and technology. After discussion, we selected a suitable problem statement for our project.

Key Learning:
We learned the importance of teamwork, discussion, problem selection, and planning before starting a project.

DAY-8 (AI VIBE CODE)

On Day 8, we learned about AI Vibe Coding, where prompts are used to guide AI in creating code. We learned how to describe our requirements clearly so that AI can generate useful code.

Key Learning:

We understood how AI can assist programmers in developing applications and solving coding problems through natural-language prompts.

DAY-9 (CUSTOMIZE)

On Day 9, we worked with the code generated using AI Vibe Coding. We learned how to run the code, check the output, identify errors, and make necessary changes. We also explored using GitHub to manage and run our project code.

Key Learning:
We learned the basic workflow of generating, testing, correcting, and managing project code.

DAY-10 (DOCUMENT)

On Day 10, we started working on website development and documentation. We learned how to organize our project information and present it in a clear and understandable format.

We also understood that documentation is important because it explains the project idea, methodology, development process, and final output.

Key Learning:
We learned how to present our technical work professionally through a website and proper documentation.

Overall Learning from Day 1 to Day 10

During the first ten days of the programme, we progressed from basic mathematical activities to practical project development. We learned how to:

- Apply mathematics to problem-solving.
- Use AI tools for learning and creativity.
- Write effective prompts.
- Work with NotebookLM and Claude AI Artifacts.
- Use AI-assisted coding techniques.
- Form a team and select a real-world problem.
- Generate and test code using AI.
- Use GitHub for project development.
- Build a basic website.

- Prepare proper project documentation.

CONCLUSION

The first 10 days of the 21-Day Programme provided us with a combination of mathematics, artificial intelligence, teamwork, coding, and project development skills. These activities helped us move from learning concepts to applying them in a practical project. The knowledge gained during these days will be useful for completing the remaining stages of our project and presenting our final work successfully.